GD vs SGD vs SVRG

Accuracy vs Gradient Computations

Setting: $\min: f(x) = \frac{1}{n}\sum_{i=1}^{n} f_i(x)$

Ex: $f(x) = \frac{1}{n}\|Ax - y\|_2^2$
$= \frac{1}{n}\sum_i (a_i^T x - y_i)^2$.

f_i is β-smooth, f is α-strongly convex

Compare: GD (GD + Momentum)
SGD
SVRG

Accuracy vs # gradient computations

GD vs SGD vs SVRG

$$f(x) = \frac{1}{n}\sum f_i(x)$$

$$\nabla f(x) = \frac{1}{n}\sum \nabla f_i(x)$$

Cost for one step of each

GD: $x_{t+1} = x_t - \eta \nabla f(x_t)$ // n gradient computations

SGD: $x_{t+1} = x_t - \eta \nabla f_I(x_t)$ // 1 gradient computation

SVRG: Recall - Outer loop: Requires a computation of full gradient, n gradient comp.

Inner loop: T SGD steps: T grad. comp.

// $n + T$ gradient computations.

We choose: $T = 10 \cdot (\beta/\alpha)$

f_i is β-smooth, f is α-s.c.

GD vs SGD vs SVRG: strongly convex and smooth

Gradient descent : $O\left((\beta/\alpha)\log(1/\varepsilon)\right)$ iterations for ε-accuracy

(using momentum) $\rightarrow O\left(\sqrt{\beta/\alpha}\log(1/\varepsilon)\right)$ " " "

$\Rightarrow O\left(\underline{n \cdot \beta/\alpha} \cdot \log(1/\varepsilon)\right)$ grad. comp. for ε accuracy.

SGD $O\left(\frac{1}{\alpha\varepsilon}\right)$ iterations for ε-accuracy.

$\Rightarrow O\left(\frac{1}{\alpha\varepsilon}\right)$ grad. computations for ε-accuracy,

SVRG : $O\left(\log(1/\varepsilon)\right)$ iterations for ε-accuracy.

$O\left(\underline{(n + \beta/\alpha)}\log(1/\varepsilon)\right)$ grad comp. for ε accuracy

GD vs SGD vs SVRG: strongly convex and smooth

SVRG is appealing when β/α is very large.

Not atypical in E.R.M. when strong convexity comes from the addition of $\|\cdot\|_2$ regularizer.

Often: $\beta = O(1)$

$\alpha = O(1/\sqrt{n})$, or $\underline{\underline{O(1/n)}}$

$\beta/\alpha \approx O(n)$